Lab 8 LCD and Touch Sensor

Objectives

1) Students should be able to display graphics using LCD.
2) Students should be able to use touch sensor.

1. Experiment

1. Screen 1: Mixing RGB colour and display using LCD.

Display the following content on LCD using the layout as shown in Fig 1.1:

- Use white background and black text.
- Show the temperature and relative humidity reading from AM2320 sensor.
- Show the horizontal scroll to adjust brightness of red, green, and blue between 0% - 100%.
- Show the colour created from the red, green, and blue mixture.
- When touching on the red, green, and blue circles, increase the brightness of each colour by 10%.
- Select the appropriate font size and adjust the layout as shown in Fig 1.1.

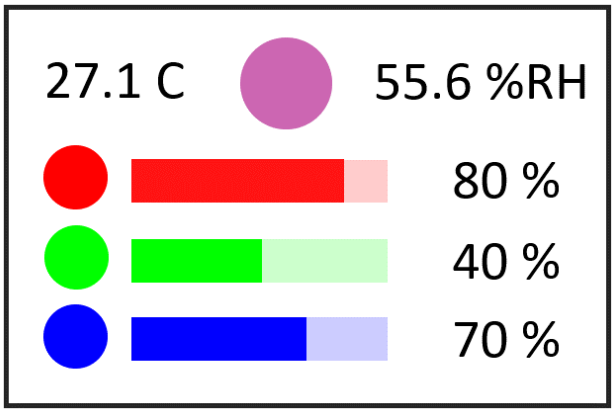


Fig 1.1 shows the screen layout for screen 1.

Special Problem 2

SP2.1. Screen 2: Display the content on LCD.

Display the following content on LCD using the layout as shown in Fig 1.2.

- Use white background and the mixied colour from Screen 1 for the text.
- Group No.
- Student's first name, last name, ID, and photo.
- Select the appropriate font size and adjust the layout as shown in Fig 1.2.

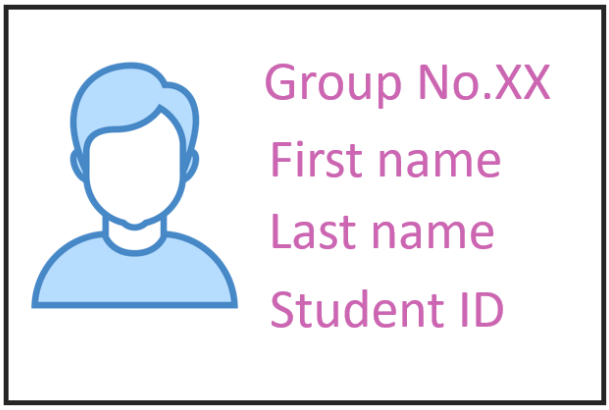


Fig 1.2 shows the screen layout for screen 2.

SP2.2. Change screens using touch sensor.

Display the **Screen 1** as shown in Fig 1.1. When touch on the mixed colour circle on top of the LCD, display the **Screen 2** as shown in Fig 1.2 for 5 seconds and then display the **Screen 1**. However, if the photo on the **Screen 2** is touched within the 5 second duration, display the **Screen 1** immediately.

Adjust LCD backlight using a potentiometer between 20% - 100%.

Lab 8 Submission

Microcontroller Interfacing 2026

Date _____ **Lab Group No.** _____

1. Student ID _____ Name _____

2. Student ID _____ Name _____

3. Student ID _____ Name _____

Checkpoints

Experiment 1